Hitec University Taxila
Department of Computer Engineering

IoT Based Smart Robotic Assistance for Alzheimer’s Patient

Group Members

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INTRODUCTION

Alzheimer’s disease affects memory, safety, and daily life activities of patients, increasing the challenges faced by caregivers and families. This project presents an IoT-Based Smart Robotic Assistance System designed to support Alzheimer’s patients through intelligent monitoring and assistance features. Using **microcontrollers, Cloud, sensors, and AI techniques**, the system aims to enhance patient safety, independence, and continuous healthcare monitoring in a smart and efficient way. Intelligent healthcare assistance for improving Alzheimer patient safety and care.

For more info scan QR Code

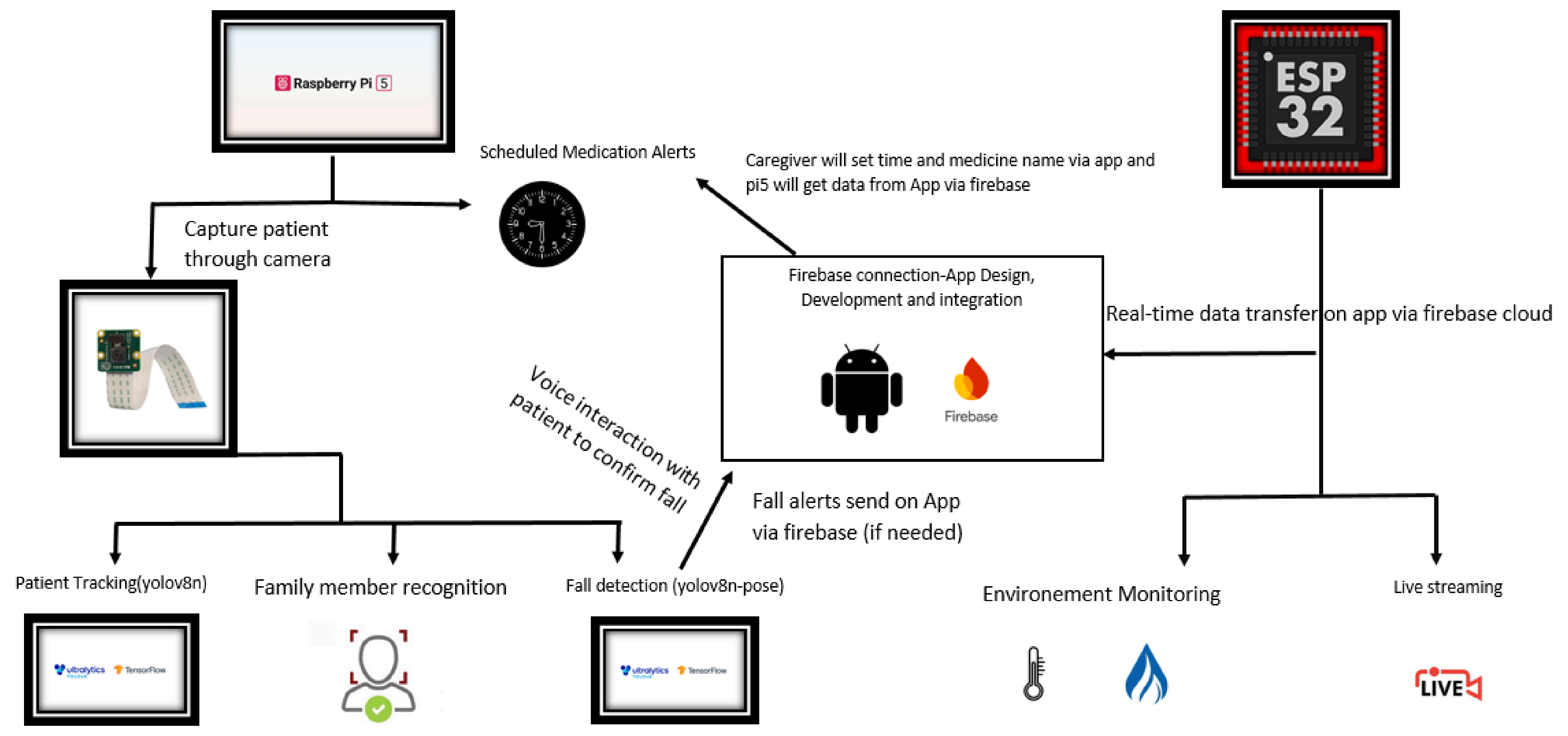
OBJECTIVES

- To detect realtime fall using YOLOv8n-pose and computer vision algorithms.
- To autonomously track/follow patient with safe-distance robot navigation for continuous monitoring.
- To perform Family member recognition using facial recognition technology.
- To generate Voice-based medication reminders through scheduled alerts through App.
- To facilitate Real-time IoT monitoring of temperature, gas, and humidity sensors.
- To enable Instant emergency notifications to caregivers via Firebase and mobile application.
- To provide live streaming of patient remotely.

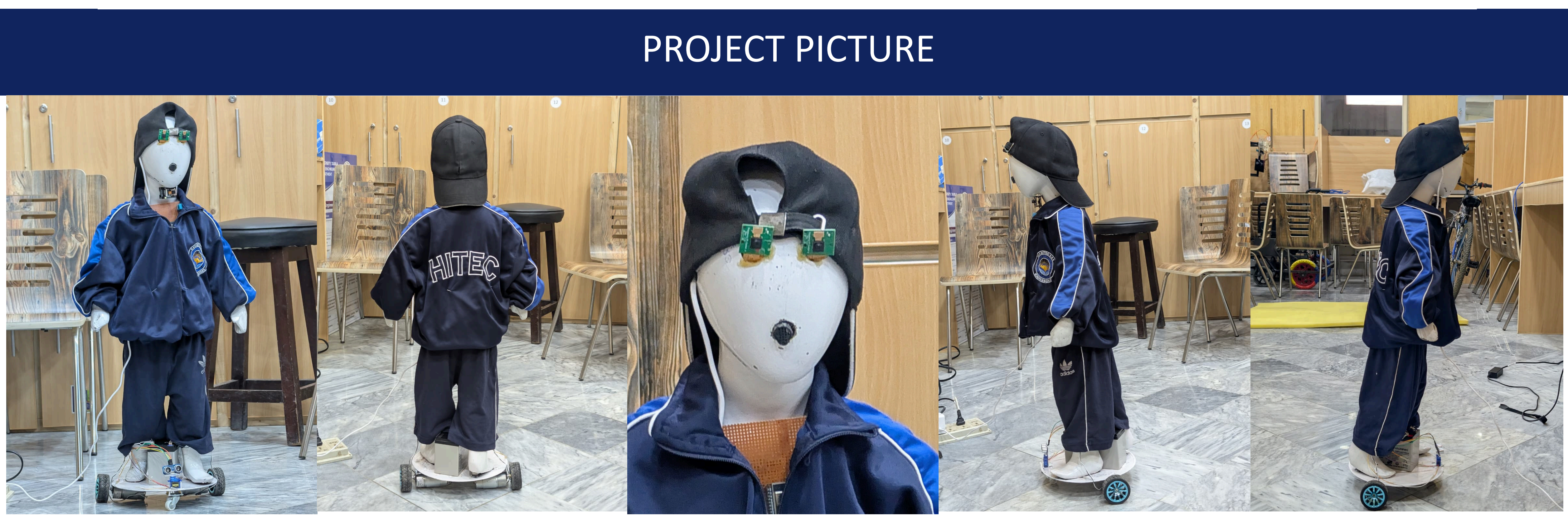
CONTRIBUTIONS

- Developed an IoT-based robotic assistant for Alzheimer patient care and monitoring, Implemented real-time fall detection, designed an autonomous patient tracking system, Integrated facial recognition to identify registered family members, developed a cloud-connected medication reminder system
- Created a mobile application for remote patient monitoring and emergency notifications
- Combined AI, IoT, cloud computing, and robotics into a single healthcare assistance platform

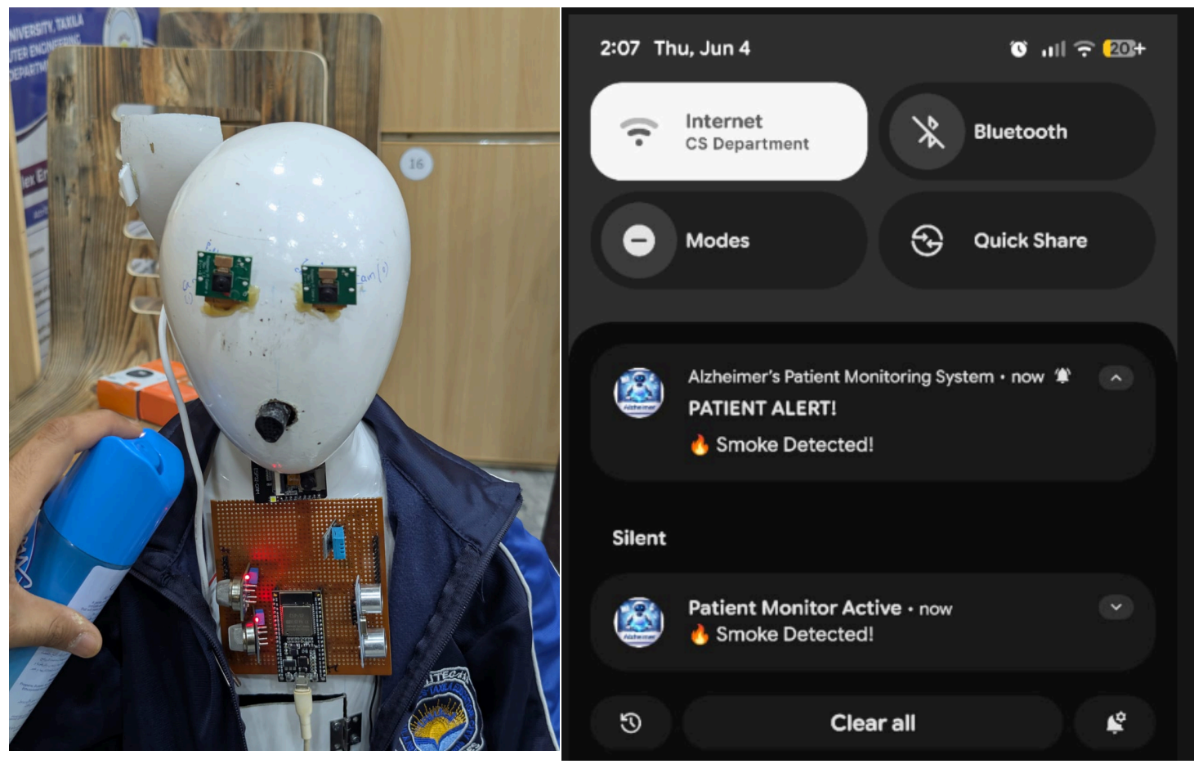
DESIGN METHODOLOGY


Raspberry Pi 5
Scheduled Medication Alerts
Caregiver will set time and medicine name via app and pi5 will get data from App via firebase
ESP 32
Capture patient through camera
Firebase connection-App Design, Development and integration
Firebase
Real-time data transfer on app via firebase cloud
Voice interaction with patient to confirm fall
Fall alerts send on App via firebase (if needed)
Patient Tracking(yolov8n)
Family member recognition
Fall detection (yolov8n-pose)
Environement Monitoring
Live streaming
LIVE

PROJECT PICTURE



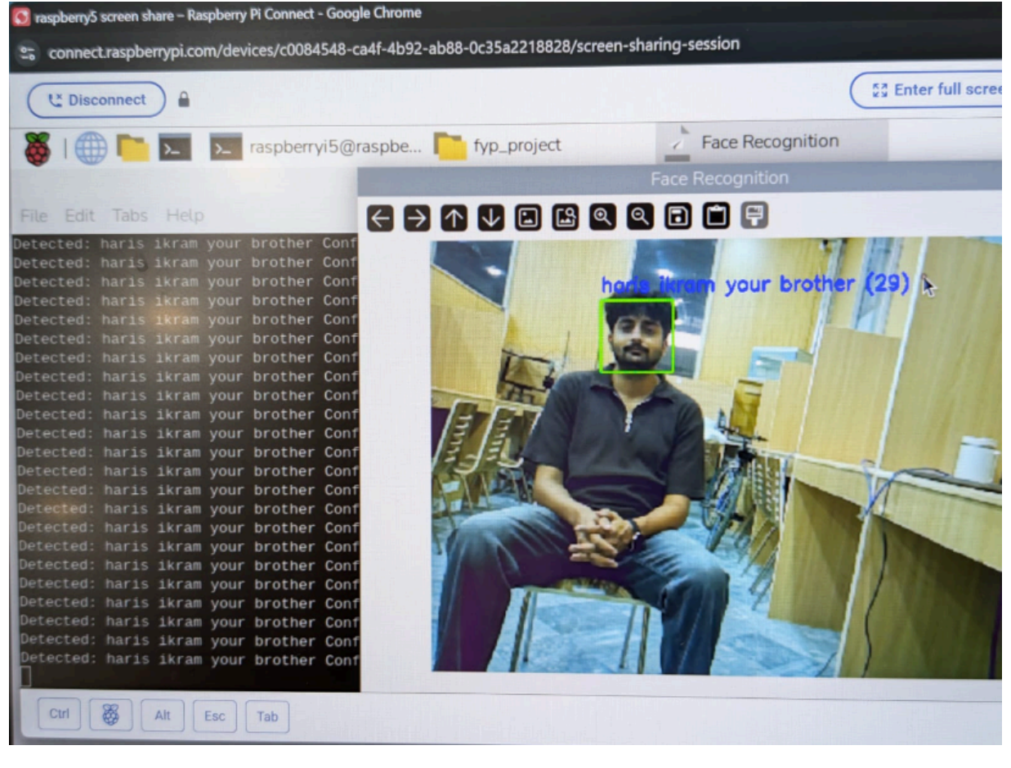
RESULTS


2:07 Thu, Jun 4
Internet
CS Department
Bluetooth
Modes
Quick Share
Silent
Clear all

Alzheimer's Patient Monitoring System • now
PATIENT ALERT!
🔥 Smoke Detected!

Patient Monitor Active • now
🔥 Smoke Detected!

Temperature,gas and smoke sensing through esp32 and sending realtime data on app


raspberry5 screen share - Raspberry Pi Connect - Google Chrome
connect.raspberrypi.com/devices/c0084548-ca4f-4b92-ab88-0c35a2218828/screen-sharing-session
Disconnect
Enter full screen
fyp_project
Face Recognition
haris ikram your brother (29)
Ctrl
Alt
Esc
Tab

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Detected: haris ikram your brother Conf
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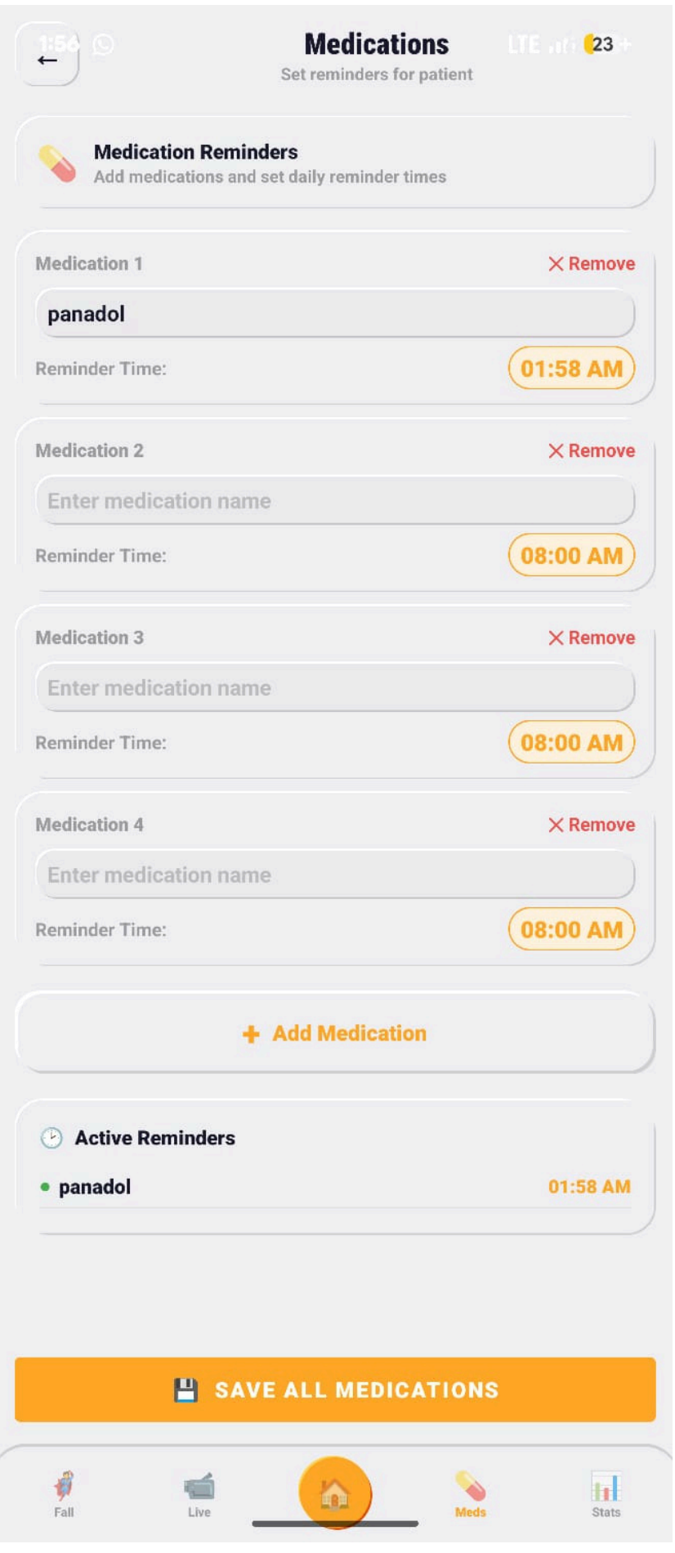
Robot will help the paitent to recognize famiy members through face Recognition and Voice assistance


Medications
Set reminders for patient
Medication Reminders
Add medications and set daily reminder times
+ Add Medication
SAVE ALL MEDICATIONS
Fall
Live
Meds
Stats

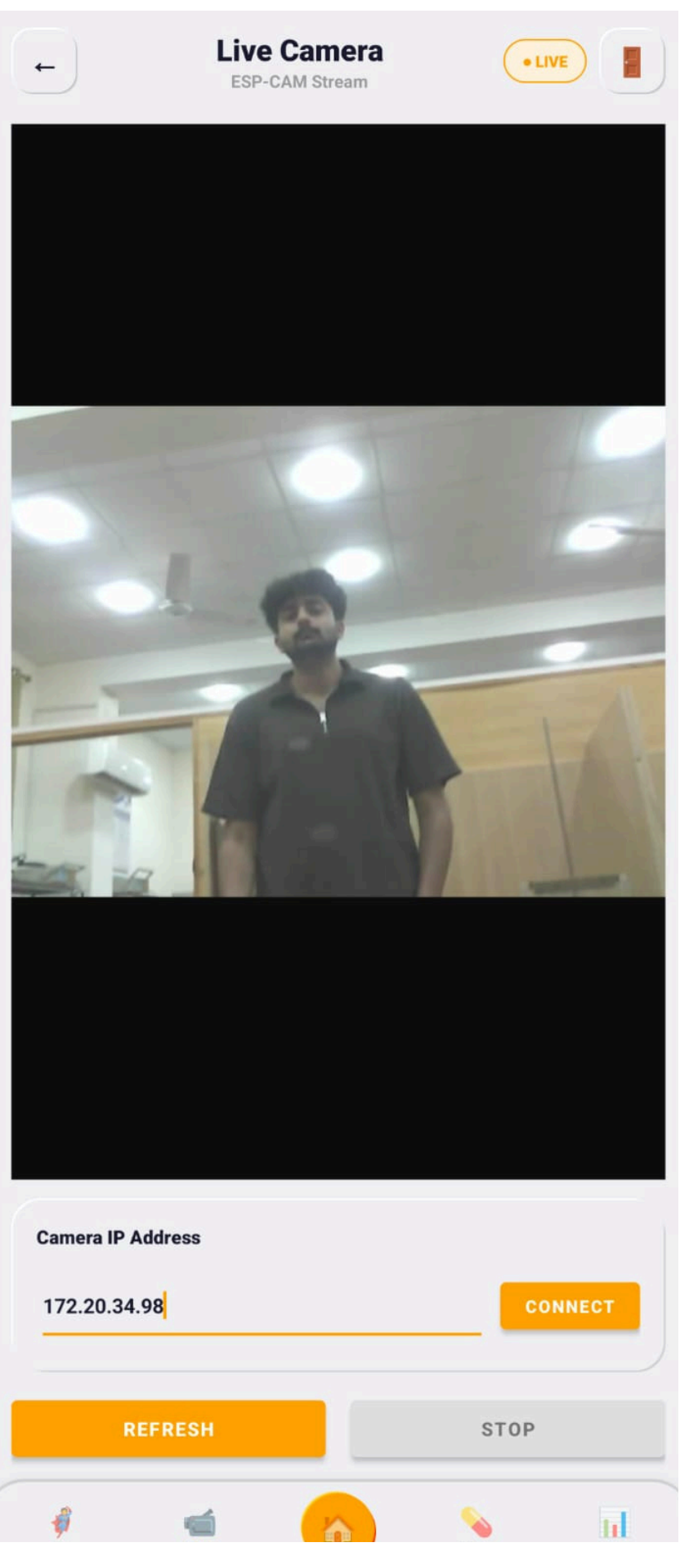
Medication 1 ✕ Remove
panadol
Reminder Time: 01:58 AM

Medication 2 ✕ Remove
Enter medication name
Reminder Time: 08:00 AM

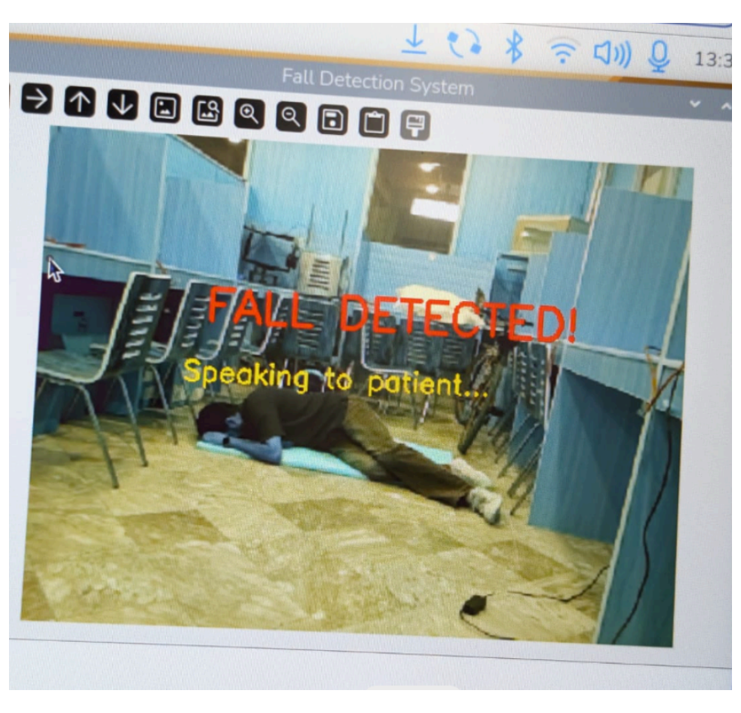
Medication 3 ✕ Remove
Enter medication name
Reminder Time: 08:00 AM

Medication 4 ✕ Remove
Enter medication name
Reminder Time: 08:00 AM

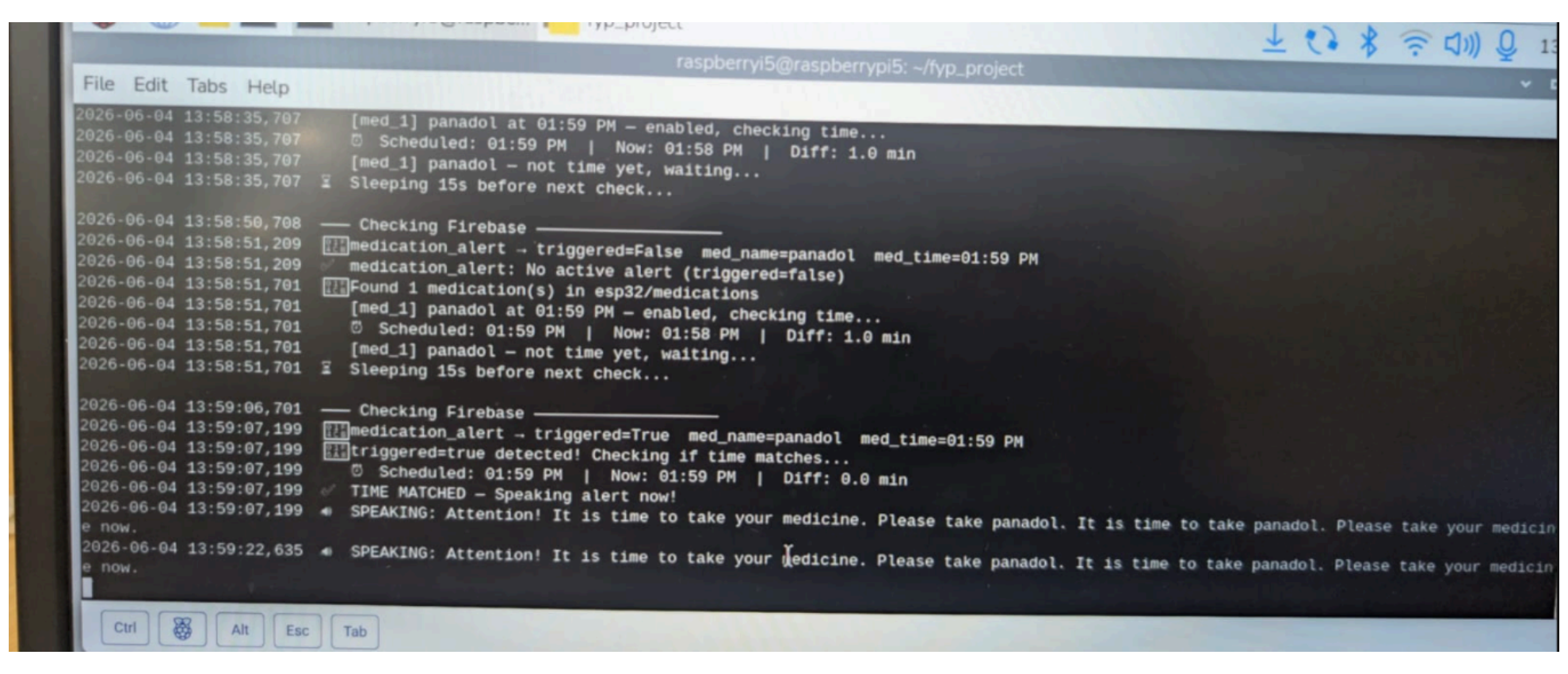
Active Reminders
• panadol 01:58 AM


Live Camera
ESP-CAM Stream
● LIVE
CONNECT
REFRESH
STOP

Camera IP Address
172.20.34.98


Fall Detection System
FALL DETECTED!
Speaking to patient...

Patient Fall Detection and Sending alert on App


raspberryi5@raspberrypi5: ~/fyp_project
File Edit Tabs Help
Ctrl
Alt
Esc
Tab

```
2026-06-04 13:58:35,707   [med_1] panadol at 01:59 PM — enabled, checking time...
2026-06-04 13:58:35,707     Scheduled: 01:59 PM  |  Now: 01:58 PM  |  Diff: 1.0 min
2026-06-04 13:58:35,707   [med_1] panadol — not time yet, waiting...
2026-06-04 13:58:35,707 ⌛ Sleeping 15s before next check...

2026-06-04 13:58:50,708 —— Checking Firebase ——————————
2026-06-04 13:58:51,209 medication_alert → triggered=False  med_name=panadol  med_time=01:59 PM
2026-06-04 13:58:51,209   medication_alert: No active alert (triggered=false)
2026-06-04 13:58:51,701 Found 1 medication(s) in esp32/medications
2026-06-04 13:58:51,701   [med_1] panadol at 01:59 PM — enabled, checking time...
2026-06-04 13:58:51,701     Scheduled: 01:59 PM  |  Now: 01:58 PM  |  Diff: 1.0 min
2026-06-04 13:58:51,701   [med_1] panadol — not time yet, waiting...
2026-06-04 13:58:51,701 ⌛ Sleeping 15s before next check...

2026-06-04 13:59:06,701 —— Checking Firebase ——————————
2026-06-04 13:59:07,199 medication_alert → triggered=True  med_name=panadol  med_time=01:59 PM
2026-06-04 13:59:07,199 triggered=true detected! Checking if time matches...
2026-06-04 13:59:07,199     Scheduled: 01:59 PM  |  Now: 01:59 PM  |  Diff: 0.0 min
2026-06-04 13:59:07,199   TIME MATCHED — Speaking alert now!
2026-06-04 13:59:07,199 ◀ SPEAKING: Attention! It is time to take your medicine. Please take panadol. It is time to take panadol. Please take your medicin
e now.
2026-06-04 13:59:22,635 ◀ SPEAKING: Attention! It is time to take your medicine. Please take panadol. It is time to take panadol. Please take your medicin
e now.
```

Generating Voice-based medication reminders through scheduled alerts through App

App is designed and developed for instant emergency notifications via firebase (the cloud)

Caregiver can see patient live footage via App.

Unit Price: 1,05,455

CONCLUSION

An IoT-Based Smart Robotic Assistance System was developed to enhance the safety, monitoring, and daily support of Alzheimer’s patients through AI and smart healthcare technologies. The robot integrates features such as fall detection, patient tracking, family member recognition, medication reminders, and real-time cloud monitoring using Raspberry Pi 5, ESP32, Firebase, and computer vision techniques. The system provides an efficient, intelligent, and cost-effective solution that assists caregivers while promoting patient independence, safety, and improved healthcare management.